

CS-301 MBSD Quiz 2 a (Q1 and Q2 mapped to CLO-1 and Q3 mapped to CLO-2).

Seat No. CS-23007

1. State whether the following statements are true or false. Give reasons for your answer. [4]
 - a. USART is half-duplex communication protocol.
 - b. Two stop bits are a must when framing the data for USART data transmission.
 - c. LSB is transferred first in USART communication protocol but we can change it to MSB using configuration registers.
 - d. There is no need to enable USART Transmission and Receive in order to use TXD and RXD pins on ATmega162.
2. With $F_{osc}=1$ MHz, find UBRR values needed to have the following Baud rates: [2]
 - a. 9600
 - b. 2400
3. Design an application using ATmega162 where a keypad is interfaced using keypad controller 74922 (inputs: rows and columns of keypad, outputs: Data available DA and DCBA (index of key pressed) that sends the ASCII value of the keypad character, when pressed, to another ATmega162 microcontroller through USART protocol i.e Pins PB3 for TXD1 and PB2 for RXD1 or Pins PD1 for TXD0 and PD0 for RXD0.. The Baud rate register value for $F_{OSC}=1$ MHz can be taken as 6. The initialization code is :
`UCSRB |= (1<<TXEN)|(1<<RXEN); //Enable receiver and transmitter`
`UCSRC |= (1<<<URSEL)|(3<<UCSZ0); // Set frame format: 8data`
 - a. Draw a neat interface (logical) diagram of the circuit. [2]
 - b. Write embedded C code for your application. [2]


```

while(1){
    keykey = getkeypressed();
    value = digit[key];
    while(!(UCSR0A & (1<<UDRE0))){ UDR = value;
        -delay-ms(100); }
}

```

```

void getkeypressed() {
    char data;
    if (PINA & (1<<PA4)) {
        data = PINA & 0x0F;
        -delay-ms(100);
        while (PINA & (1<<PA4)) {
            return data; } } }

```

Receiver:-

```

DDRA |= (1<<PD1);
DDRB
DDRB |= ~(1<<PD0);
UBRRH = 0x00;
UBR0L = '0x06;
UCSRB |= (1<<RXEN);
UCSR |= (1<<URSEL) | (3<<UCSZ0);
char receive-data;
DPR = 0xFF;
while(1){
    while(!(UCSR0A & (1<<RXIF))){
        receive-data = UDR0;
        -delay-ms(100); }
    PORTC = receive-data;
}

```

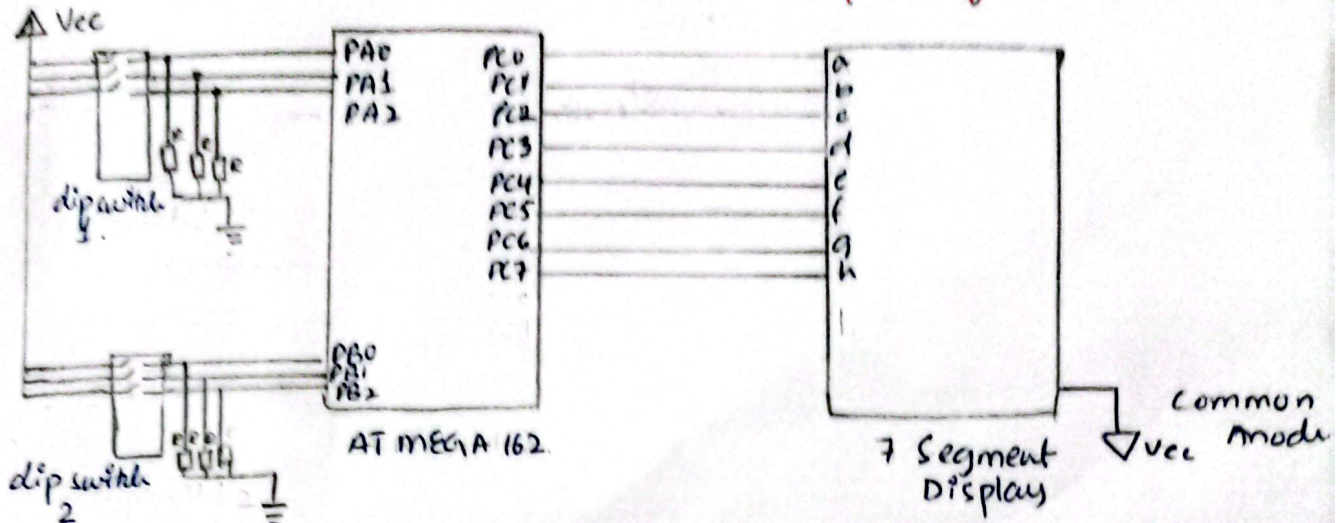

Design an ATmega162 based circuit to take two 3-bit operands as input from the DIP switches, add the two operands and display the sum in its equivalent hexadecimal value on seven-segment display without using BCD decoder.

- Make a clear interface for your application. [5]
- Write an embedded C code according to your interface diagram. [5]

QUIZ

February 17, 2026

10/10 Points



10 = [
 11 =]
 12 = □
 13 =
 14 = E

(5)

```
(6) #include <avr/io.h>
#define F_CPU 1000000UL
#include <util/delay.h>
int main(void)
{
    DDRA = 0b00000000;
    DDRB = 0b00000000;
    DDRC = 0b11111111;

    while(1)
    {
        value = (PINA & 0b00000111) + (PINB & 0b00000111);
        if (value == 0b00000000)
            PORTC = 0b11000000;
        else if (value == 0b00000001)
            PORTC = 0b10111100;
        else if (value == 0b00000010)
            PORTC = 0b10100100;
        else if (value == 0b00000011)
            PORTC = 0b10110000;
        else if (value == 0b00000100)
            PORTC = 0b10011001;
        else if (value == 0b00000101)
            PORTC = 0b10010010;
        else if (value == 0b00000110)
            PORTC = 0b10001101;
        else if (value == 0b00000111)
            PORTC = 0b10000110;
    }
}
```

```
else if (value == 0b00000110)
    PORTC = 0b10000110;
else if (value == 0b00000111)
    PORTC = 0b11111000;
else if (value == 0b00001000)
    PORTC = 0b10000000;
else if (value == 0b00001001)
    PORTC = 0b10011000;
else if (value == 0b00001010)
    PORTC = 0b10010100;
else if (value == 0b00001011)
    PORTC = 0b10000011;
else if (value == 0b00001100)
    PORTC = 0b11000110;
else if (value == 0b00001101)
    PORTC = 0b11000000;
else if (value == 0b00001110)
    PORTC = 0b10000110;
else if (value == 0b00001111)
    PORTC = 0b10000110;
}
```